

Assurance Boundaries for Post-Quantum X.509

Abstract

Post-quantum certificate material changes certificate size, parser stress, path-building behavior, transport limits, and validation assumptions. This manuscript organizes assurance around reproducible cases: corpus entries, validator decisions, parser behavior, and lint reports.

Methodology

The harness builds certificate-path cases with classical, hybrid, and post-quantum material. Each case is parsed, linted, minimized where useful, and checked against path-validation behavior. The paper records which assumptions move when certificate size, algorithm identifiers, and chain composition change.

Results

The current results isolate assurance boundaries for post-quantum X.509: parser stress, transport limits, path-building assumptions, and regression cases that should remain visible during migration.